

Requirement Analysis Phase

1. Purpose of the Phase

Requirement Analysis turns the problems found in the Ideation Phase into a precise statement of what the Courier Management System must do and how well it must do it. It covers the customer journey, user roles, functional and non-functional requirements, data flow, technology choices, and a traceability matrix that links every requirement to the part of the system that delivers it and the test that proves it.

2. Customer Journey Map

The journey map follows a parcel from booking to reporting. For each stage it records what the user does, what frustrates them today, and how CMS responds.

Stage	Activity	Pain point today	CMS response
Booking	Customer asks the branch to send a parcel	Details written on paper; phone numbers often wrong	Customer record with 10-digit phone and 6-digit pincode validation
Shipment creation	Staff records addresses, weight and price	Zero or wrong weights lead to wrong charges	Shipment record with auto-numbered tracking ID and weight and price validation
Agent assignment	Dispatcher chooses an agent	Assignment by memory or phone; double booking	Assigned Agent lookup showing agent status and branch
Transit	Parcel travels to the destination	No record of movement	Delivery Update records with date, status and remarks
Delivery update	Agent reports progress	Status reaches the customer late	Flow moves the shipment to In Transit automatically
Invoice	Customer is billed	Bills prepared separately from shipment data	Invoice linked to the shipment with payment status
Reporting	Manager reviews results	Data collected manually	Reports and Courier Management Dashboard

3. User Roles

Role	Responsibilities	Main objects used
Courier Admin	Configuration, security, full data control	All objects
Branch Manager	Books shipments, assigns agents, raises invoices, reviews dashboards	Customer, Shipment, Delivery Agent, Branch, Delivery Update, Invoice
Delivery Agent	Delivers parcels and posts status updates	Shipment, Delivery Update
Customer Support	Handles customer queries and updates contact details	Customer, Shipment, Delivery Update

4. Solution Requirements

4.1 Functional requirements

FR	Requirement	Description	Priority	Delivered by
FR-1	Customer management	Create, view and edit customers with name, email, phone, address, city, state, pincode, type and active flag. Phone must be 10 digits and pincode 6 digits.	Must	Customer object
FR-2	Shipment booking	Book a shipment with addresses, weight, price, type, expected date and status. Weight and price must be positive and the expected date cannot be in the past.	Must	Shipment object
FR-3	Delivery agent assignment	Maintain agents with branch and status, and assign one agent to each shipment.	Must	Delivery Agent, Branch
FR-4	Real-time status updates	Record each tracking event with status, date and remarks; the shipment status updates automatically. A Delayed event requires remarks.	Must	Delivery Update, Flow
FR-5	Invoice generation	Create an invoice per shipment with amount, date and payment status.	Must	Invoice object
FR-6	Reports and dashboards	Show shipment status, agent performance and revenue by branch.	Should	Reports, Dashboard

4.2 Non-functional requirements

NFR	Description	How it is met
Usability	Easy Lightning interface	One app, one tab per object, sectioned page layouts
Security	Role-based access control	Four profiles and four roles; object and tab permissions
Reliability	Error-free automation	Declarative validation rules and a tested record-triggered flow
Performance	Fast tracking updates	Flow updates one parent record in the same transaction
Availability	24/7 cloud access	Hosted on the Salesforce cloud; no local servers
Scalability	Supports growing branches	New branches and agents are added as records

5. Use Cases

UC	Use case	Actor	Main flow
UC-1	Register customer	Branch Manager	Open Customers, click New, enter details, Save. Invalid phone or pincode is blocked.
UC-2	Book shipment	Branch Manager	Open Shipments, select customer, enter addresses, weight, price, expected date, Save.
UC-3	Assign agent	Branch Manager	Open shipment, set Assigned Agent, Save.
UC-4	Post delivery update	Delivery Agent	Open Delivery Updates, New, choose shipment and status, Save. Shipment status changes to In Transit.
UC-5	Raise invoice	Branch Manager	Open Invoices, New, choose shipment, enter amount and status, Save.
UC-6	View dashboard	Branch Manager	Open Dashboards, choose Courier Management Dashboard, click Refresh.

6. Data Requirements

Object	Data held
Customer	Name, email, phone, address, city, state, pincode, customer type, active flag
Shipment	Tracking number, customer, source and destination address, weight, price, status, expected date, assigned agent, shipment type
Delivery Agent	Name, email, phone, branch, status, joining date
Branch	Name, code, city, state, manager name
Delivery Update	Update number, shipment, status, update date, remarks
Invoice	Invoice number, shipment, amount, invoice date, payment status

7. Data Flow

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

Step	Data movement
Customer input	Customer details are entered and saved to the Customer object after validation
Shipment creation	A Shipment record references the Customer and receives an auto-numbered tracking ID
Validation	Validation rules check weight, price, date, phone and pincode before saving
Agent assignment	The Assigned Agent lookup links the shipment to a Delivery Agent
Delivery update	Delivery Update records are created; the flow updates the parent shipment
Invoice	An Invoice record references the shipment
Reports	Reports group shipments, agents and revenue; the dashboard displays the results

8. Technology Stack

Layer	Technology	Reason
UI	Salesforce Lightning	Ready-made responsive interface, apps and tabs
Logic	Salesforce Flows	Point-and-click automation without code
Database	Custom objects	Structured data with relationships and history tracking
Automation	Validation rules, flows	Data quality and status updates
Security	Profiles, roles	Access control by job function
Reporting	Reports, dashboards	Built-in analytics with no extra tools

9. Requirement Traceability Matrix

FR	Requirement	Object / component	Validation or automation	Test case
FR-1	Customer management	Customer	Phone, pincode, email rules	TC-1
FR-2	Shipment booking	Shipment	Weight, price, date rules	TC-2
FR-3	Agent assignment	Delivery Agent, Branch	Agent email, branch code rules	Configuration review
FR-4	Status updates	Delivery Update	Status flow, delay remarks rule	TC-3
FR-5	Invoice generation	Invoice	Lookup to Shipment	TC-4

FR	Requirement	Object / component	Validation or automation	Test case
FR-6	Reports and dashboards	Reports, dashboard	Grouped reports	TC-5

10. Constraints and Assumptions

- The system runs in a Salesforce Developer Edition org with a limited number of user licenses.
- Customers do not log in; branch staff enter and read data on their behalf.
- Notifications, GPS tracking and payment collection are outside this release.
- Invoice amounts are entered by staff; there is no automatic tax or discount calculation.